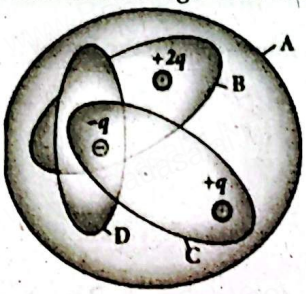
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12 - STD	QUARTERLY EXAMINATION - 2026	
TIME :3.00 Hrs	PHYSICS	MARKS : 70

PART – I

Note: Answer all the questions, choose the correct answer. **15 X 1 = 15**

1. Four Gaussian surfaces are given below with charges inside each Gaussian surface. Rank the electric flux through each Gaussian surface in increasing order.



(a) D < C < B < A (b) A < B = C < D

(c) C < A = B < D (d) D > C > B > A

2. An electric field $\vec{E} = 10x\hat{\imath}$ exists in a certain region of space. Then the potential difference $V=V_0-V_A$, where V_0 is the potential at the origin and V_A is the potential at $x = 2$ m is
(a) 10V (b) –20 V (c) +20 V (d) –10 V
3. If the critical angle of the water in a fish tank is 48.6°, then the angle of vision of the fish is
(a) 45° (b) 48.6° (c) 90° (d) 97.2°
4. In India electricity is supplied for domestic use at 220 V. It is supplied at 110 V in USA. If the resistance of a 60W bulb for use in India is R, the resistance of a 60W bulb for use in USA will be
(a) R (b) 2R (c) $\frac{R}{4}$ (d) $\frac{R}{2}$
5. The internal resistance of a 2.1 V cell which gives a current of 0.2 A through a resistance of 10Ω is
(a) 0.2 (b) 0.5 Ω (c) 0.8 Ω (d) 1.0 Ω
6. A circular coil of radius 5 cm and 50 turns carry a current of 3 ampere. The magnetic dipole moment of the coil is
(a) 1.0 A m² (b) 1.2 A m² (c) 0.5 A m² (d) 0.8 A m²
7. A wire of length l carries a current I along the Y direction and magnetic field is given by $\vec{B} = \frac{B}{\sqrt{3}}(\hat{\imath}+\hat{\jmath}+\hat{k})$T, The magnitude of Lorentz force acting on the wire is
(a) $\sqrt{\frac{2}{3}}\beta Il$ (b) $\sqrt{\frac{1}{3}}\beta Il$ (c) $\sqrt{2}\beta Il$ (d) $\sqrt{\frac{1}{2}}\beta Il$
8. The flux linked with a coil at any instant t is given by $\phi_B = 10t^2-50t+250$ The induced emf at t=2s is
(a) –190 V (b) –10 V (c) 10 V (d) 190 V
9. In a transformer, the number of turns in the primary and the secondary are 410 and 1230 respectively. If the current in primary is 6A, then that in the secondary coil is
(a) 2 A (b) 18 A (c) 12 A (d) 1 A
10. Which of the following magnetic materials possesses negative magnetic susceptibility?
(a) Dia (b) Para (c) Ferro (d) Para and Ferro
11. Let E = Eo sin [10^6x -ωt] be the electric field of plane electromagnetic wave, the value of ω is
(a) 0.3 × 10^{-14} rad s^{-1} (b) 3 × 10^{-14} rad s^{-1}
(c) 0.3 × 10^{14} rad s^{-1} (d) 3 × 10^{14} rad s^{-1}
12. If the magnetic monopole exists, then which of the Maxwell's equation to be modified?
(a) $\oint_s \vec{E}.\overrightarrow{dA} = \frac{Q_{closed}}{\varepsilon_0}$ (b) $\oint \vec{B}.\overrightarrow{dA} = 0$
(c) $\oint \vec{B}.\overrightarrow{dl} = \mu_0 I_c + \mu_0 \varepsilon_0 \frac{d}{dt}\oint_s \vec{E}.\overrightarrow{dA}$ (d) $\oint \vec{B}.\overrightarrow{dl} = -\frac{d\phi B}{dt}$

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13. A ray of light travelling in a transparent medium of refractive index n falls, on a surface separating the medium from air at an angle of incidents of 45^0. The ray can undergo total internal reflection for the following n,

(a) n = 1.25 (b) n = 1.33 (c) n = 1.4 (d) n = 1.5

14. If the velocity and wavelength of light in air is V_a and λ_a and that in water is V_w and λ_w, then the refractive index of water is,

(a) $\frac{V_w}{\lambda_a}$ (b) $\frac{V_a}{V_w}$ (c) $\frac{\lambda_w}{\lambda_a}$ (d) $\frac{V_a}{V_w}\frac{\lambda_a}{\lambda_w}$

15. When a dielectric is inserted into a capacitor while it is disconnected from the battery:

(a) Capacitance decreases, energy increases
(b) Capacitance increases, electric field increases
(c) Capacitance increases, energy decreases
(d) Capacitance decreases, electric field increases

Part - II

Note: i) Answer any six questions.
ii) Q.No. 24 is compulsory. **6 × 2 = 12**

16. State Gauss's law.
17. What is the Peltier effect?
18. How can the current sensitivity of a galvanometer be increased?
19. State Lenz's law.
20. What is displacement current?
21. State two uses of ultraviolet rays.
22. What are paraxial rays and marginal rays?
23. A spherical stone and a spherical metal ball of the same size and mass are dropped from the same height. Which one a stone or a metal ball will reach the Earth's surface first? justify your answer. Assume there is no air friction.
24. Find the speed of light traveling through pure water which has a refractive index of 1.33.

Part - III

Note: i) Answer any six questions.
ii) Q.No. 33 is compulsory. **6 × 3 = 18**

25. Derive the expression for the equivalent capacitance when capacitors are connected in series.
26. What is the Seebeck effect? What are the applications of the Seebeck effect?
27. State and explain the principle of a potentiometer.
28. Mention the characteristics of the magnetic Lorentz force.
29. Explain why alternating current (AC) is superior to direct current (DC).
30. What are Fraunhofer lines? How do they help in identifying the elements present in the Sun?
31. Derive the equation for the effective focal length of lenses kept in contact with each other.
32. What are critical angle and total internal reflection?
33. Calculate the electric flux passing through a rectangle with sides of 5cm and 10cm placed in a region of uniform electric field with a magnitude of 100 N C^{-1}. The given angle is $\theta = 60°$. What would the electric flux be if θ were zero?

Part - IV

Note: Answer all the questions. **5 × 5 = 25**

34. a) Calculate the electric field at a point on the equatorial plane of an electric dipole. (Or)
b) (i) State Coulomb's law. (ii) State the differences between Coulomb force and gravitational force.
35. a) Derive the condition for the balanced state of a Wheatstone bridge. (Or)
b) Describe the microscopic model of electric current and derive the microscopic form of Ohm's law from it.
36. a) Explain in detail the working of cyclotron. (or)
b) Calculate the force acting on a current carrying conductor placed in a magnetic field.
37. a) Describe the principle, construction, and working of a transformer. (Or)
b) Calculate the current, total impedance, and phase angle in a series RLC circuit.
38. a) Explain the modification made by Maxwell to Ampere's circuital law. (Or)
b) What is dispersion? Derive an expression for the dispersive power of a medium.

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